

Beauty Clinic Website

Phase 2: System Analysis & Design

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SYSTEM ANALYSIS

01

Stakeholders & Actors

The Beauty Clinic Website system has two primary actors who interact with the system in distinct roles. Each actor has a defined set of responsibilities and system access privileges.

User (Client)

Registered or guest client who visits the website

- Register / Login to the system
- Browse available services
- View clinic specialists
- Book appointments
- View booking status and history
- Browse skincare products
- Place product orders
- Send messages to the clinic
- View before & after gallery

Admin

Clinic staff member with full system management access

- Manage services (Add / Edit / Delete)
- Manage specialists profiles
- Manage skincare products
- View all booking requests
- Approve or reject bookings
- View client messages
- Monitor dashboard statistics
- Manage gallery content

SYSTEM ANALYSIS

02 System Requirements

Functional Requirements

ID	Requirement	Actor	Priority
FR-01	The system shall allow users to register and log in using email and password.	User	High
FR-02	The system shall display all available clinic services with name, price, and duration.	User	High
FR-03	The system shall allow users to view specialist profiles.	User	Medium
FR-04	The system shall allow logged-in users to book an appointment by selecting service, specialist, date, and time.	User	High
FR-05	The system shall save new bookings with status "Pending" until reviewed by admin.	System	High
FR-06	The system shall allow admin to approve or reject appointment requests.	Admin	High
FR-07	The system shall notify users of booking status changes.	System	Medium
FR-08	The system shall allow users to browse and purchase skincare products.	User	Medium
FR-09	The system shall allow users to send contact messages to the clinic.	User	Medium
FR-10	The admin shall be able to manage services, specialists, and products (CRUD).	Admin	High
FR-11	The system shall display a before & after gallery of treatment results.	User	Low

Non-Functional Requirements

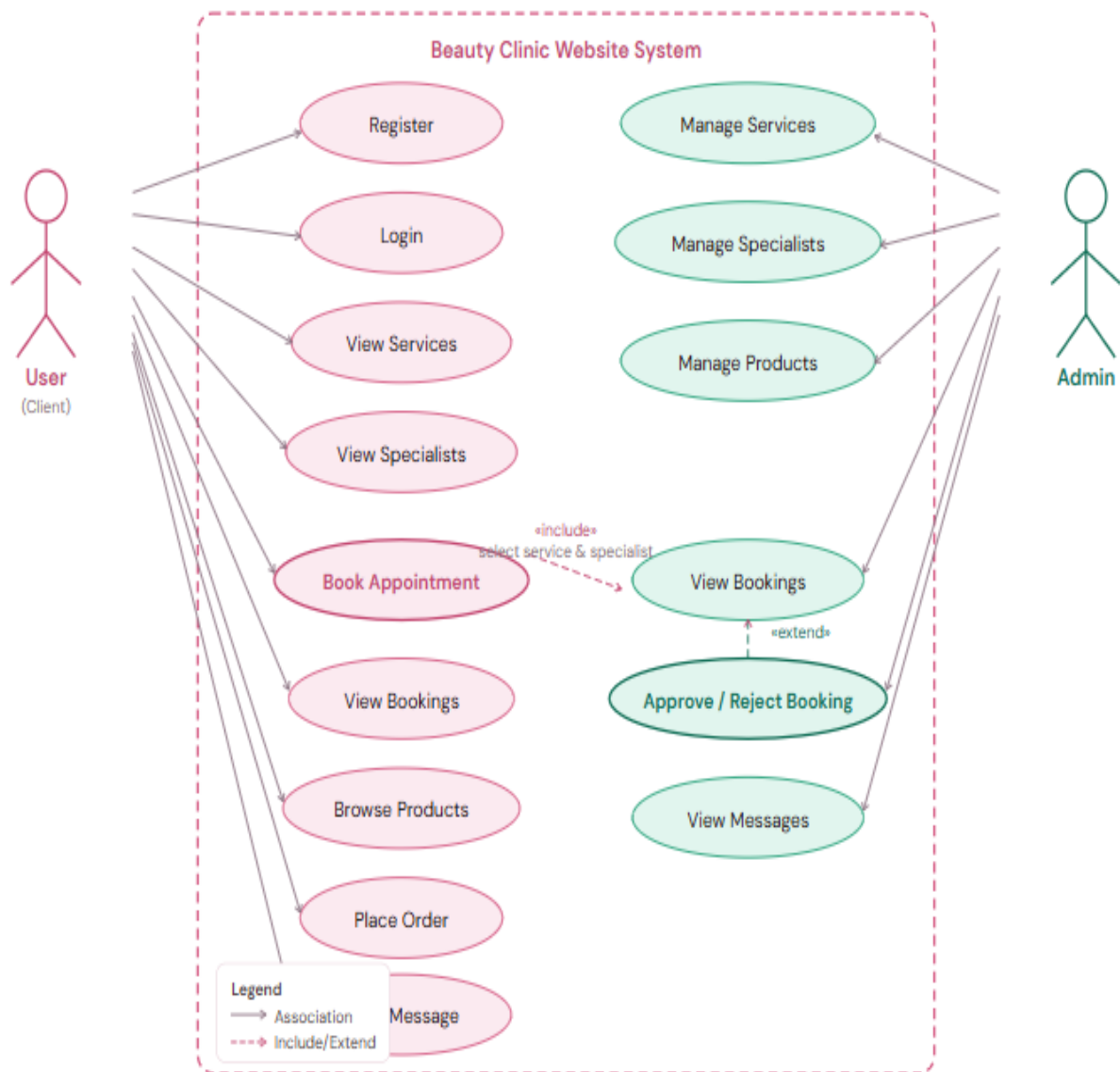
ID	Category	Requirement
NFR-01	Performance	Pages must load within 3 seconds under normal conditions.
NFR-02	Usability	The system interface must be easy to navigate for non-technical users.
NFR-03	Security	User passwords must be hashed before storage. Sessions must be authenticated.
NFR-04	Reliability	The system must maintain data integrity; no booking data should be lost.
NFR-05	Scalability	The database design must support growth in users and bookings over time.
NFR-06	Compatibility	The website must be responsive and functional on mobile, tablet, and desktop.
NFR-07	Maintainability	Code must be modular and well-commented for ease of future updates.

SYSTEM DESIGN

03 Use Case Diagram

The following Use Case Diagram illustrates all interactions between the two primary actors (User and Admin) and the Beauty Clinic Website system.

UML USE CASE DIAGRAM — BEAUTY CLINIC WEB SITE



SYSTEM DESIGN

04 Use Case Description

The following is a detailed description of the most critical use case in the system: **Book Appointment**.

Use Case Name	Book Appointment
Use Case ID	UC-05
Actor	User (Client)
Description	Allow a registered and logged-in user to submit an appointment request by selecting a service, a specialist, a preferred date, and a time slot. The request is saved as "Pending" until an admin reviews and confirms or rejects it.
Preconditions	• The user is registered in the system. • The user is logged in. • At least one service and one specialist are available in the system.
Main Flow	1. The user navigates to the "Book Appointment" page. 2. The system displays a booking form with available services. 3. The user selects a service from the dropdown list. 4. The system updates the form to show available specialists for the chosen service. 5. The user selects a specialist. 6. The user selects a preferred date from the date picker. 7. The user selects a preferred time slot. 8. The user clicks the "Submit Booking" button. 9. The system validates all fields are filled. 10. The system saves the booking record in the database with status = "Pending". 11. The system displays a confirmation message to the user: "Your appointment request has been submitted successfully." 12. The admin is notified of the new pending request in the dashboard.
Alternative Flow (Admin Rejects)	▪ After step 12, the admin reviews the booking request in the Admin Dashboard. ▪ The admin selects "Reject" for the booking. ▪ The system updates the booking status to "Rejected". ▪ The system notifies the user that their appointment request was rejected. ▪ The user may submit a new booking request with different details.
Alternative Flow (Validation Error)	At step 9, if any field is empty or invalid, the system displays an error message. The system highlights the missing or invalid fields. The use case returns to step 3 for the user to correct the input.
Postconditions	• A new booking record exists in the database with status "Pending", "Confirmed", or "Rejected". • The user can view their booking in "My Bookings" page. • The admin can see the request in the Admin Dashboard.

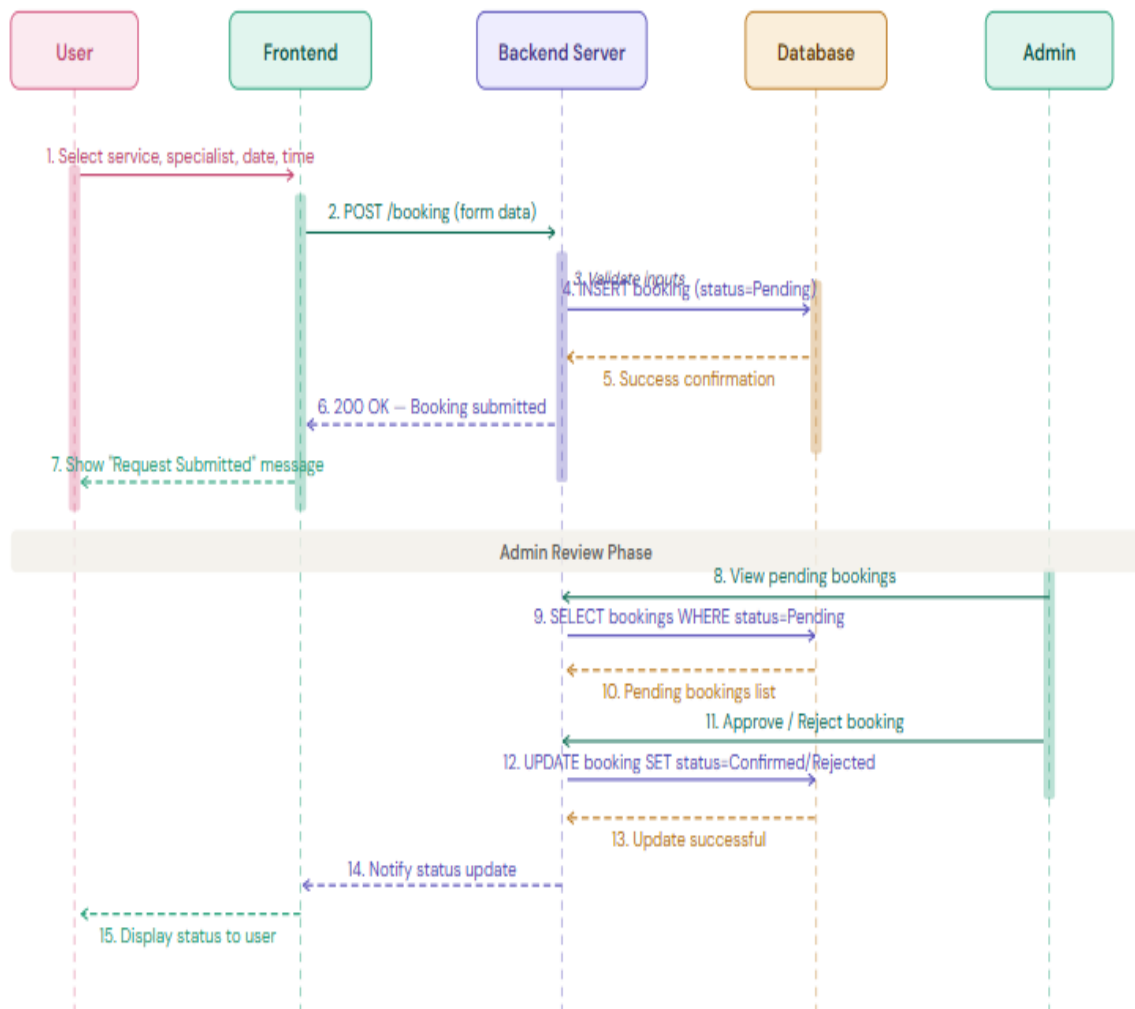
SYSTEM DESIGN

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Sequence Diagram — Book Appointment

This sequence diagram illustrates the interaction between objects during the "Book Appointment" use case, from the moment the user selects a service to the admin's approval or rejection.

SEQUENCE DIAGRAM — BOOK APPOINTMENT



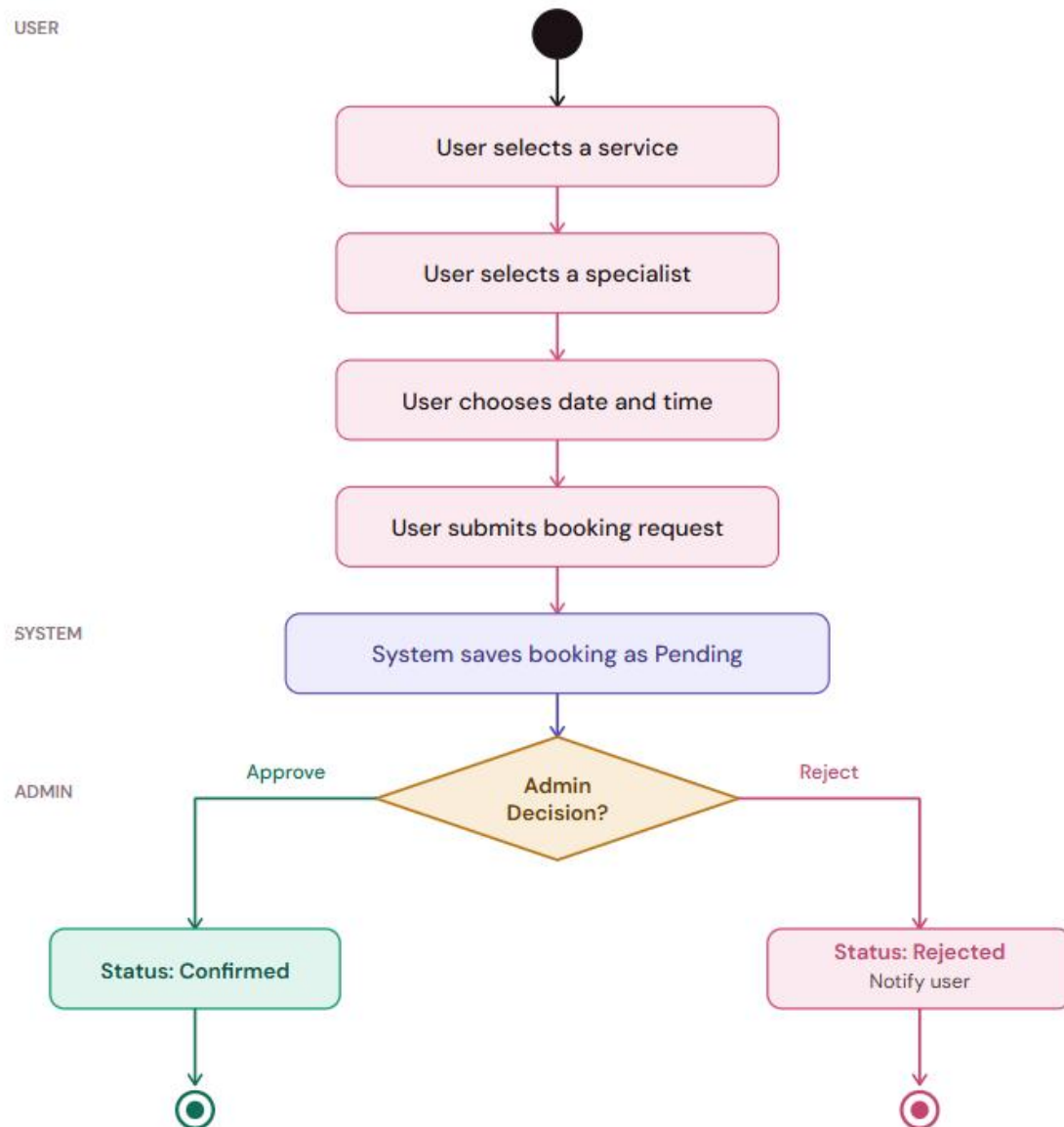
SYSTEM DESIGN

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Activity Diagram — Booking Process

The activity diagram below shows the complete flow of the booking process, including the admin decision point with branches for approval and rejection.

ACTIVITY DIAGRAM — BOOK APPOINTMENT

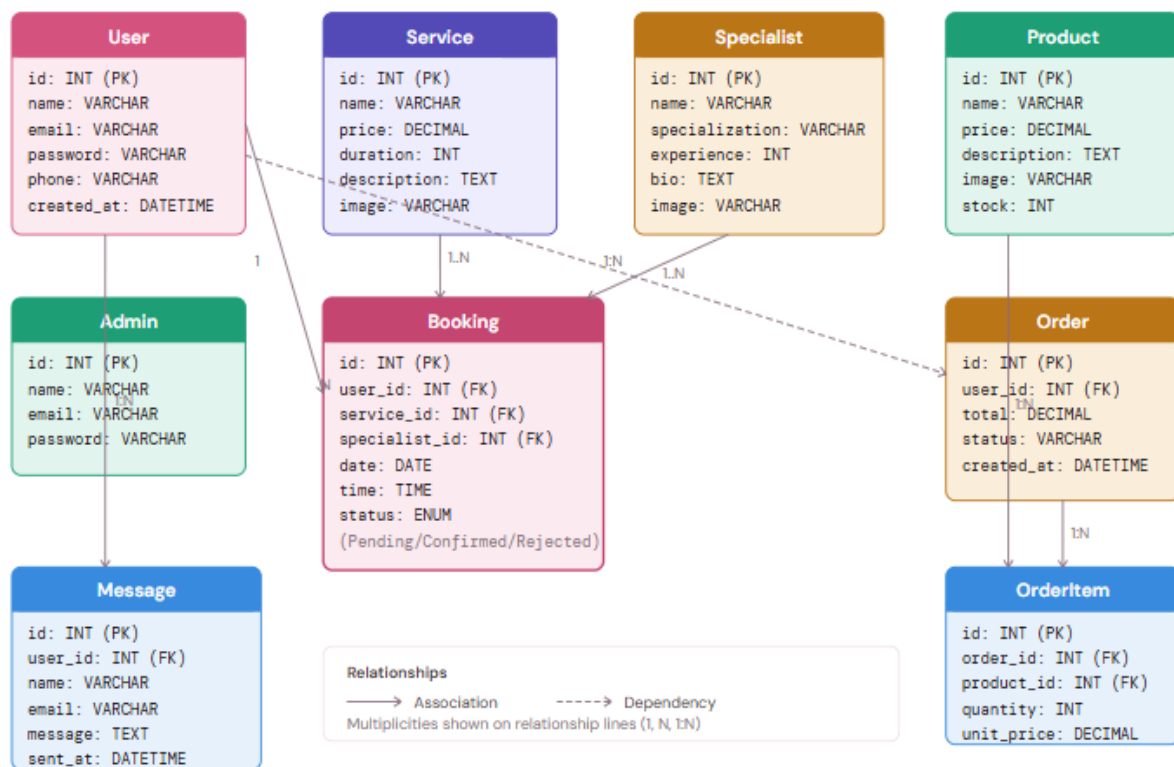


SYSTEM DESIGN

Class Diagram

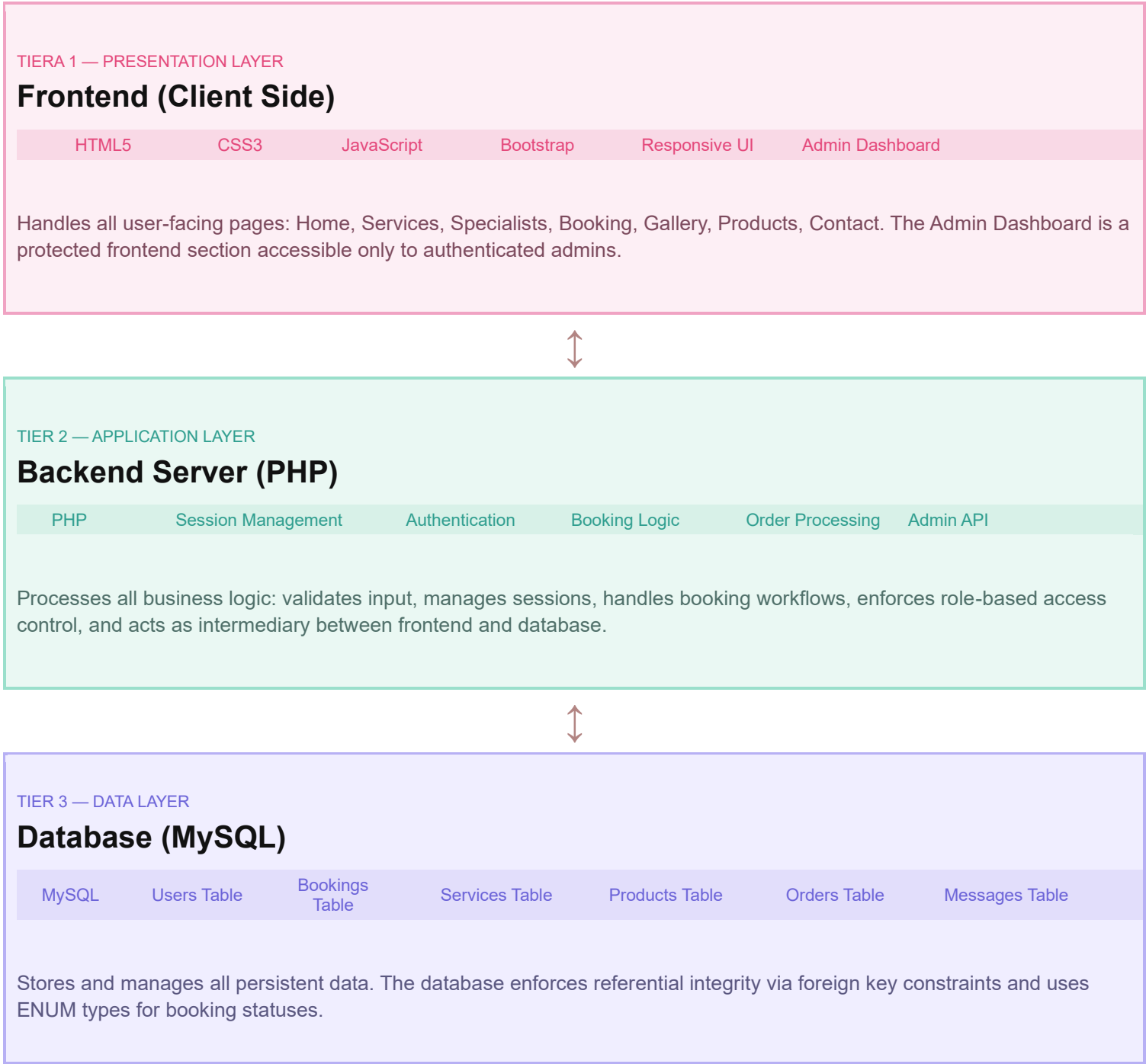
The class diagram shows the main system classes, their attributes, and the relationships (associations and multiplicities) between them.

UML CLASS DIAGRAM – BEAUTY CLINIC WEBSITE



High-Level System Architecture

The Beauty Clinic Website follows a classic 3-Tier Architecture, separating the presentation, logic, and data layers. This ensures modularity, maintainability, and scalability.

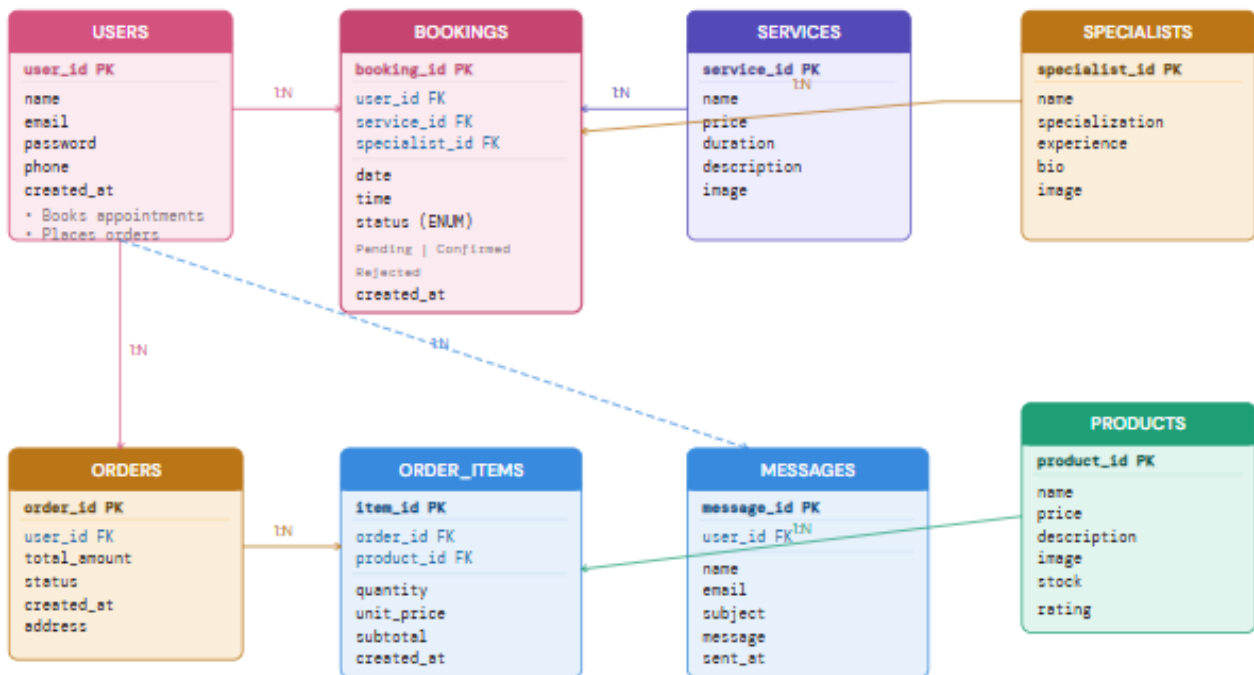


SYSTEM DESIGN

Entity Relationship Diagram (ERD)

The ERD shows all database entities, their primary keys, foreign key relationships, and cardinalities. All relationships are 1:N (one-to-many).

ERD — BEAUTY CLINIC WEBSITE DATABASE



Legend
 PK = Primary Key FK = Foreign Key —→ 1:N Relationship

SYSTEM DESIGN

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Database Schema

The data dictionary provides a field-level description of every attribute in the database, including data type, constraints, and purpose.

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Users:

FIELD	TYPE	CONSTRAINTS	DESCRIPTION
user_id PK	INT	AUTO_INCREMENT, NOT NULL	Unique identifier for each registered user
name	VARCHAR(100)	NOT NULL	Full name of the user
email	VARCHAR(150)	NOT NULL, UNIQUE	Email address used for login and communication
password	VARCHAR(255)	NOT NULL	Hashed password for authentication
phone	VARCHAR(20)	Nullable	Optional contact phone number
created_at	DATETIME	DEFAULT NOW()	Timestamp of account creation

Services:

FIELD	TYPE	CONSTRAINTS	DESCRIPTION
service_id PK	INT	AUTO_INCREMENT	Unique identifier for each clinic service
name	VARCHAR(100)	NOT NULL	Name of the beauty treatment (e.g., Botox, Laser)
price	DECIMAL(10,2)	NOT NULL	Service price in USD
duration	INT	Nullable	Session duration in minutes
description	TEXT	Nullable	Detailed description of the treatment
image	VARCHAR(255)	Nullable	Path to service image file

Specialists:

FIELD	TYPE	CONSTRAINTS	DESCRIPTION
specialist_id PK	INT	AUTO_INCREMENT	Unique identifier for each doctor or specialist
name	VARCHAR(100)	NOT NULL	Full name of the specialist
specialization	VARCHAR(100)	Nullable	Area of medical expertise
experience	INT	Nullable	Years of professional experience
bio	TEXT	Nullable	Professional biography shown on the Specialists page
image	VARCHAR(255)	Nullable	Path to specialist profile photo

Bookings:

FIELD	TYPE	CONSTRAINTS	DESCRIPTION
booking_id PK	INT	AUTO_INCREMENT	Unique identifier for each appointment request
user_id FK	INT	NOT NULL → users	References the user who made the booking
service_id FK	INT	NOT NULL → services	References the selected service
specialist_id FK	INT	NOT NULL → specialists	References the selected specialist/doctor
date	DATE	NOT NULL	Requested appointment date
time	TIME	NOT NULL	Requested appointment time slot
status	ENUM	DEFAULT 'Pending'	Booking lifecycle state: Pending, Confirmed, or Rejected
created_at	DATETIME	DEFAULT NOW()	Timestamp when the booking was submitted

Products:

FIELD	TYPE	CONSTRAINTS	DESCRIPTION
product_id PK	INT	AUTO_INCREMENT	Unique identifier for each skincare product
name	VARCHAR(100)	NOT NULL	Product name (e.g., Vitamin C Serum, Sunscreen)
price	DECIMAL(10,2)	NOT NULL	Product sale price in USD
description	TEXT	Nullable	Product description and ingredients
image	VARCHAR(255)	Nullable	Path to product image file
stock	INT	DEFAULT 0	Available inventory quantity
rating	DECIMAL(2,1)	DEFAULT 0.0	Average customer rating (0.0 – 5.0)

Orders:

FIELD	TYPE	CONSTRAINTS	DESCRIPTION
order_id PK	INT	AUTO_INCREMENT	Unique identifier for each customer order
user_id FK	INT	NOT NULL → users	References the user who placed the order
total_amount	DECIMAL(10,2)	NOT NULL	Total price of all items in the order
status	VARCHAR(50)	DEFAULT 'Pending'	Order fulfillment status (Pending, Shipped, Delivered)
address	TEXT	Nullable	Shipping/delivery address
created_at	DATETIME	DEFAULT NOW()	Timestamp when the order was placed

order_items:

FIELD	TYPE	CONSTRAINTS	DESCRIPTION
item_id PK	INT	AUTO_INCREMENT	Unique identifier for each line item
order_id FK	INT	NOT NULL → orders	References the parent order
product_id FK	INT	NOT NULL → products	References the purchased product
quantity	INT	NOT NULL, DEFAULT 1	Number of units ordered
unit_price	DECIMAL(10,2)	NOT NULL	Price per unit at time of purchase (snapshot)
subtotal	DECIMAL(10,2)	GENERATED (quantity * unit_price)	Computed total cost for this line item

Messages:

FIELD	TYPE	CONSTRAINTS	DESCRIPTION
message_id PK	INT	AUTO_INCREMENT	Unique identifier for each contact message
user_id FK	INT	Nullable → users	References registered user; NULL for guest messages
name	VARCHAR(100)	NOT NULL	Sender's full name (from contact form)
email	VARCHAR(150)	NOT NULL	Sender's email address for reply
subject	VARCHAR(200)	Nullable	Message subject line
message	TEXT	NOT NULL	Full message body text
sent_at	DATETIME	DEFAULT NOW()	Timestamp when the message was submitted
is_read	TINYINT(1)	DEFAULT 0	Admin read status: 0 = unread, 1 = read